Chapter 13 Conservation of energy

The Energy in the World

Thermal Energy

Electrical Energy

Atomic Energy

Chemical Energy

Mechanical Energy

$$\Delta(K+U)=0$$

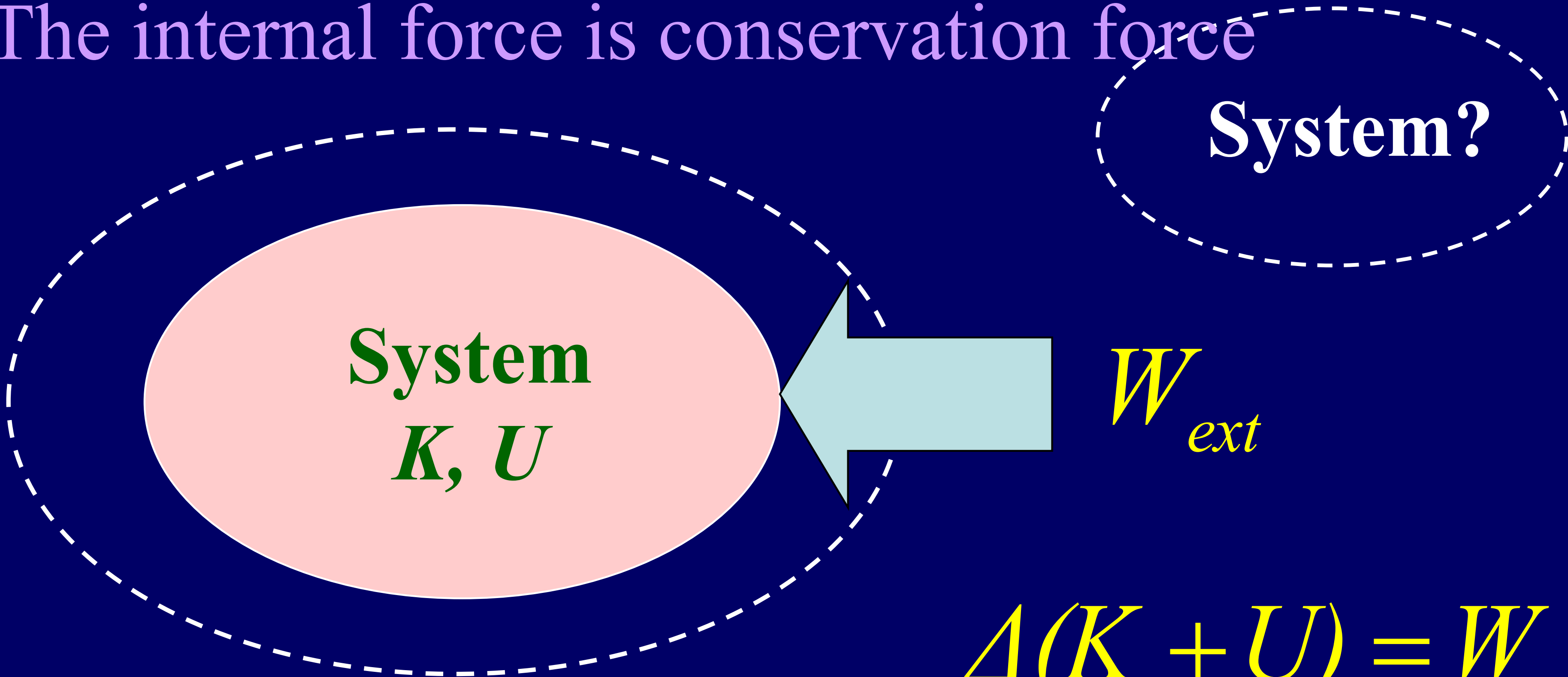
$\left\{ \begin{array}{l} \text{Kinetic Energy} \\ \text{Potential Energy} \end{array} \right.$

No work done on a system by external forces

The internal force is conservation force

Work done on a system by external forces

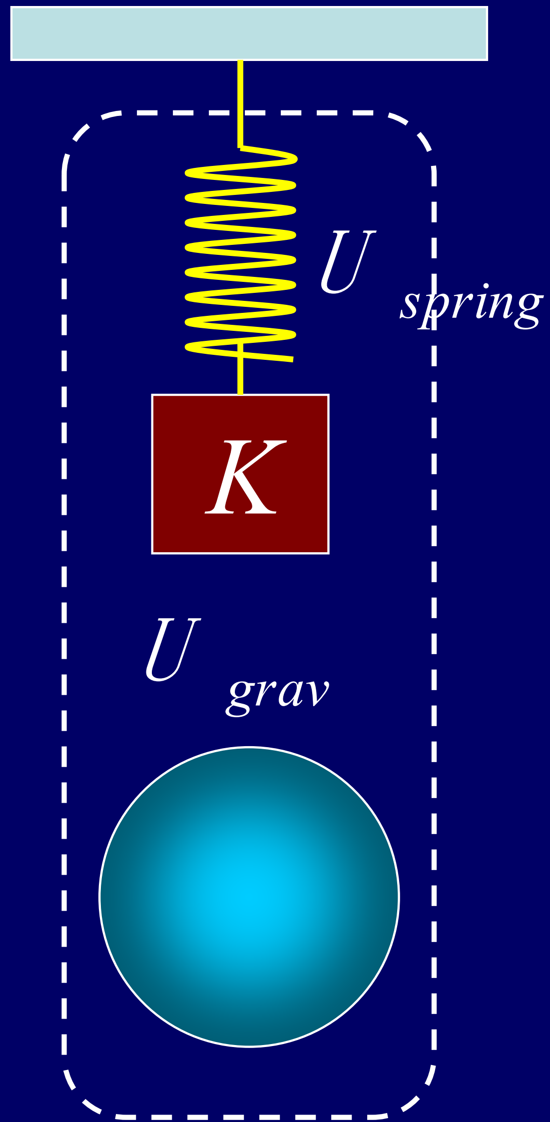
The internal force is conservation force



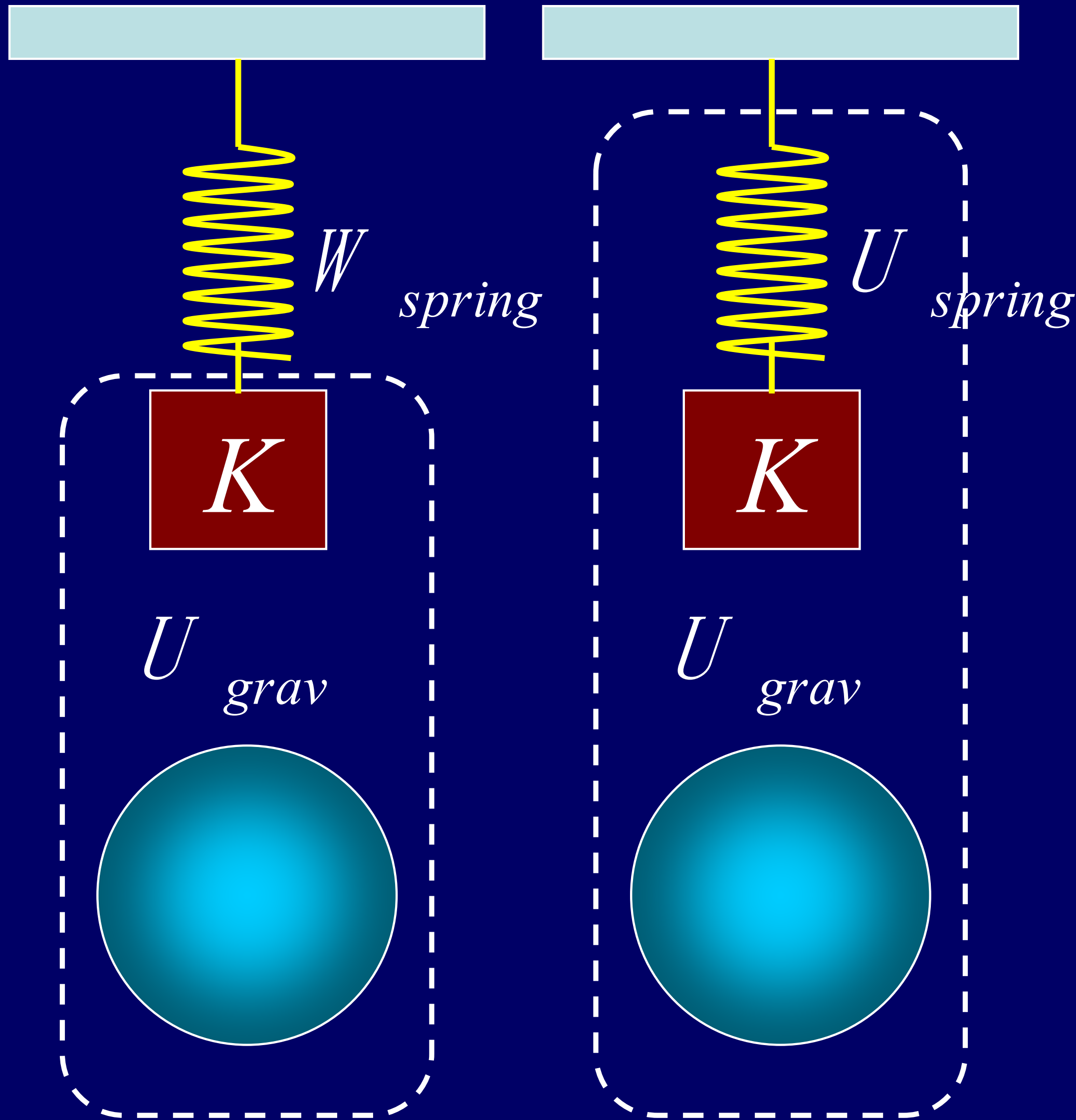
$$\Delta(K+U)=W_{ext}$$

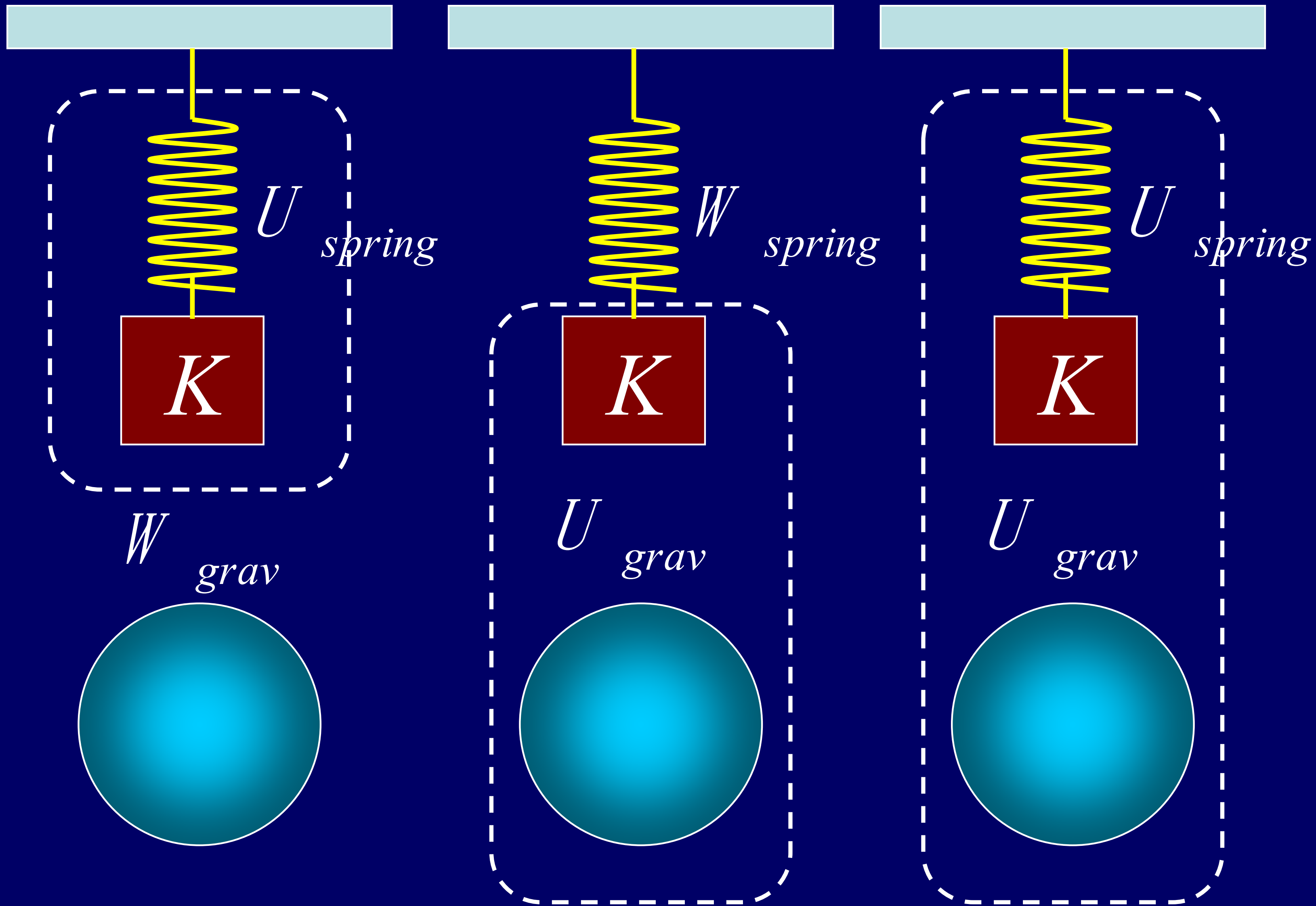
$$W_{ext} = 0$$

$$\Delta U_{spring} + \Delta U_{grav} + \Delta K = 0$$

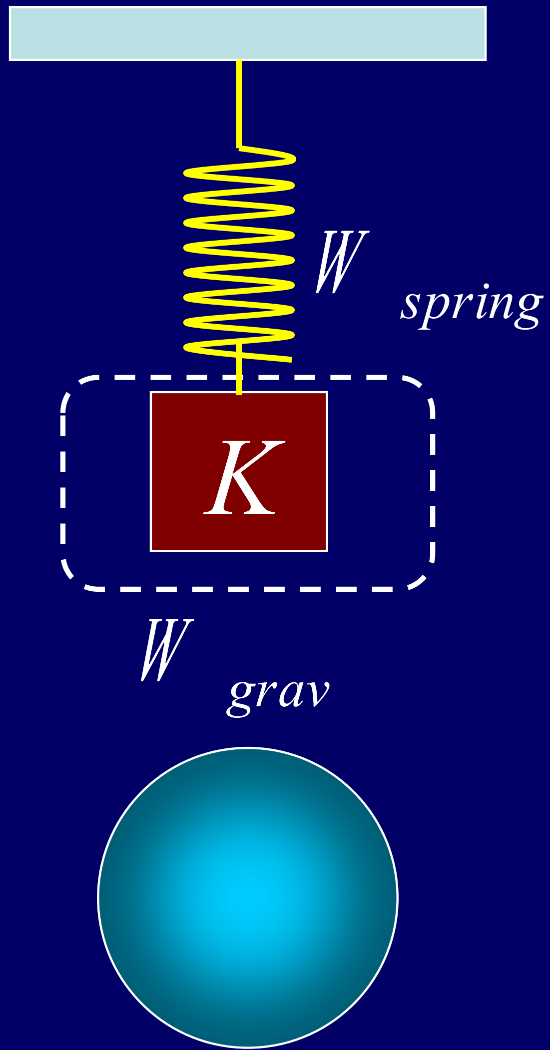


$$W_{spring} = \Delta U_{grav} + \Delta K$$





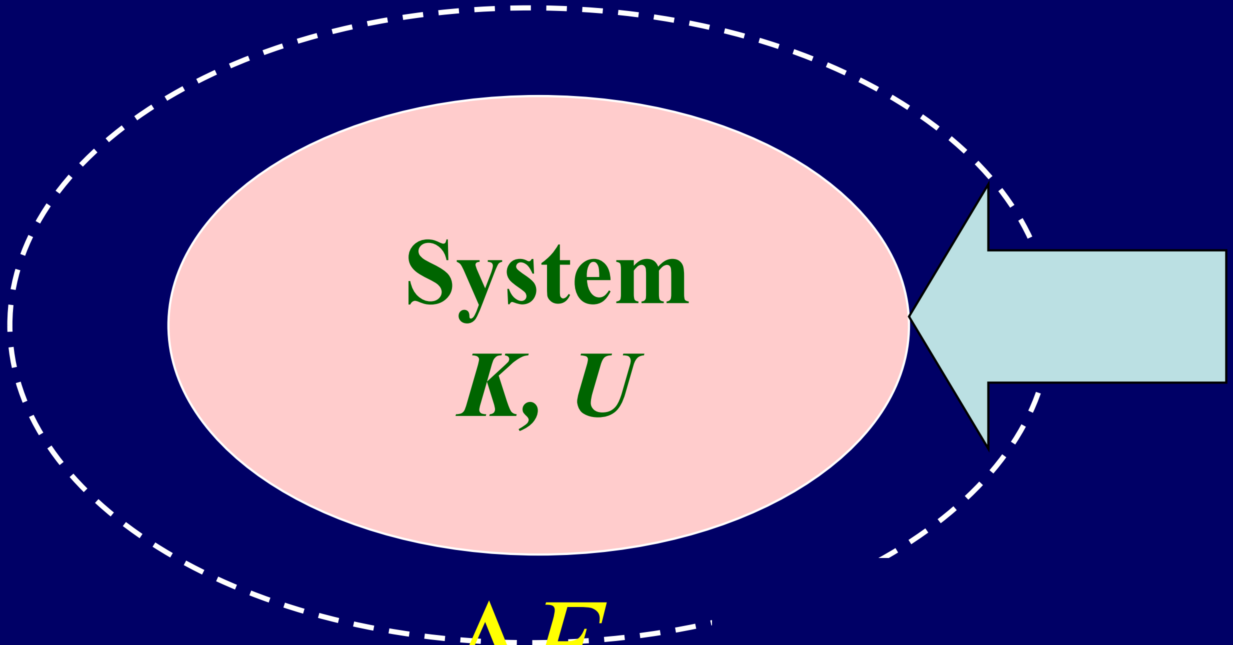
$$W_{grav} = \Delta U_{spring} + \Delta K$$



$$W_{grav} + W_{spring} = \Delta K$$

Work done on a system by external forces

The internal force **is not** conservation force



W_{ext} Q

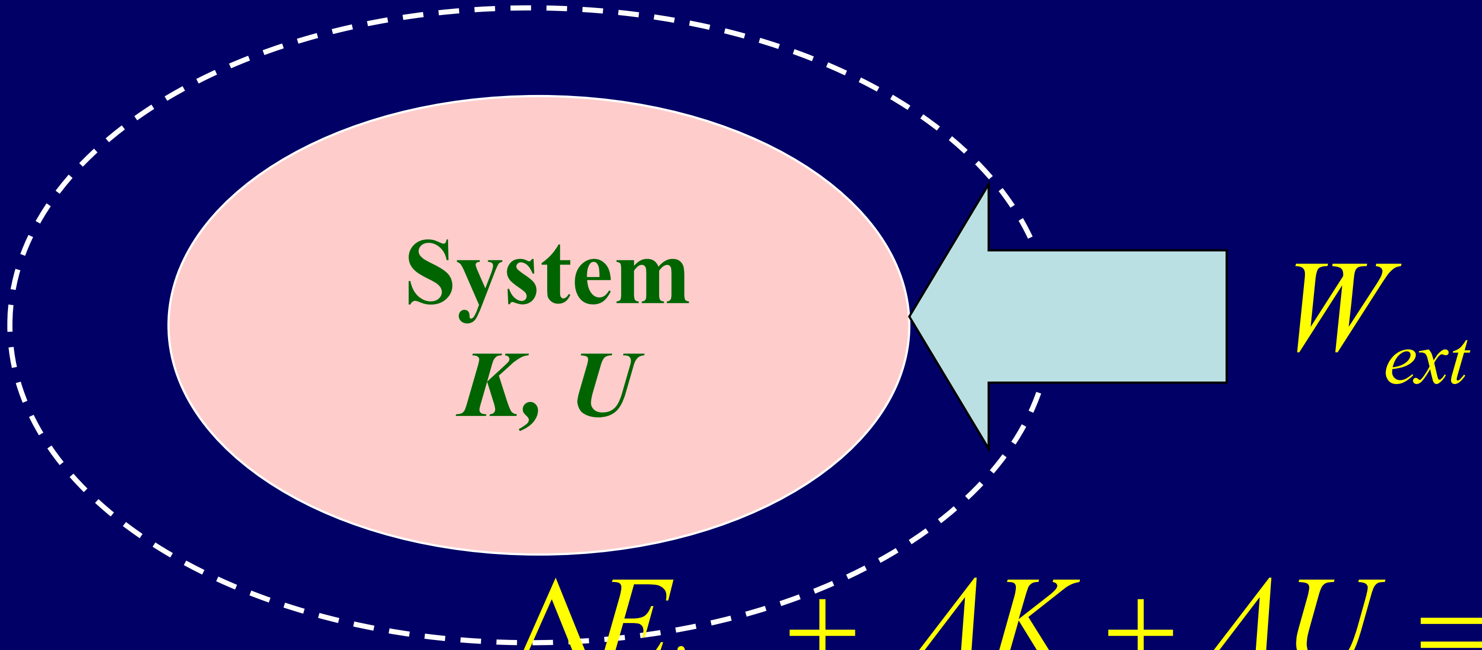
$$\Delta E_{int} = W_{ext} + Q$$

$$\Delta E_{int} = W_{ext} + Q$$ Mechanical Energy

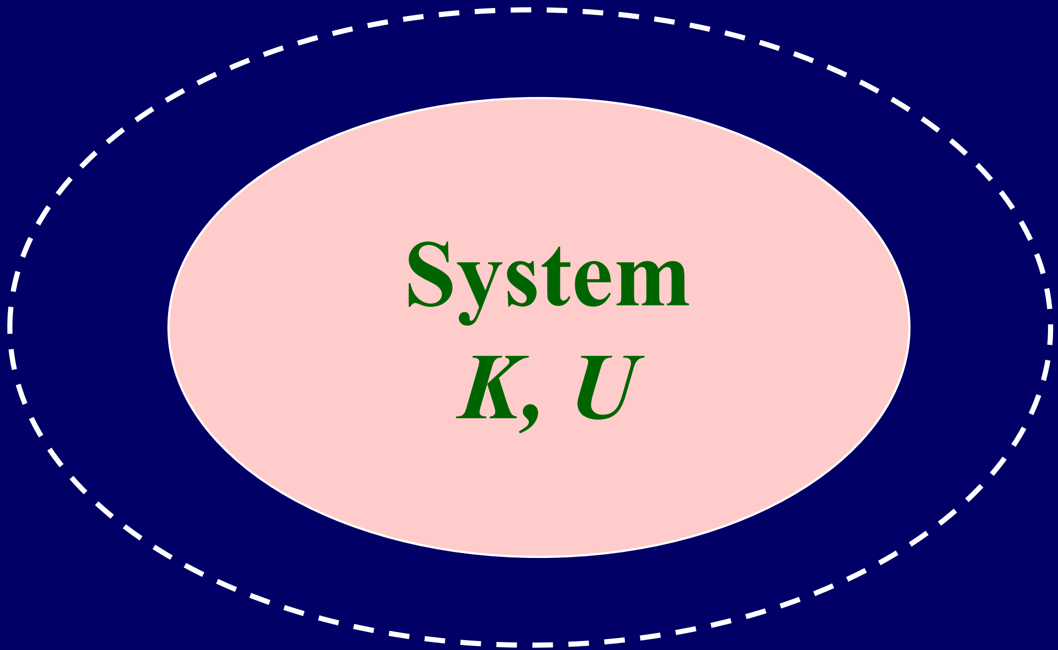
The first law of thermodynamics

Work done on a system by external forces

The internal force **is not** conservation force



$$\Delta E_{\text{int}} + \underbrace{\Delta K + \Delta U}_{\text{Mechanical Energy}} = W_{ext}$$



The center of mass

$$\vec{r}_{cm} = \frac{1}{M}\sum m_i \vec{r}_i$$

$$\vec{v}_{cm} = \frac{1}{M}\sum m_i \vec{v}_i$$

$$\vec{a}_{cm} = \frac{1}{M}\sum m_i \vec{a}_i$$

The momentum of system

$$M\vec{v}_{cm} = \sum m_i \vec{v}_i = \sum \vec{p}_i$$

The force on system

$$M\vec{a}_{cm} = \sum m_i \vec{a}_i = \sum \vec{F}_i$$

The energy on system?

The energy on system

$$\vec{F}_{ext} = M\vec{a}_{cm} \qquad F_{ext}dx_{cm} = M\frac{dv_{cm}}{dt}dx_{cm} = Mv_{cm}dv_{cm}$$

$$W_{ext} \neq \int_{x_i}^{x_f} F_{ext}dx_{cm} = \int_{v_{cm,i}}^{v_{cm,f}} Mv_{cm}dv_{cm} = \frac{1}{2}Mv_{cm,f}^2 - \frac{1}{2}Mv_{cm,i}^2 = \Delta K_{cm}$$

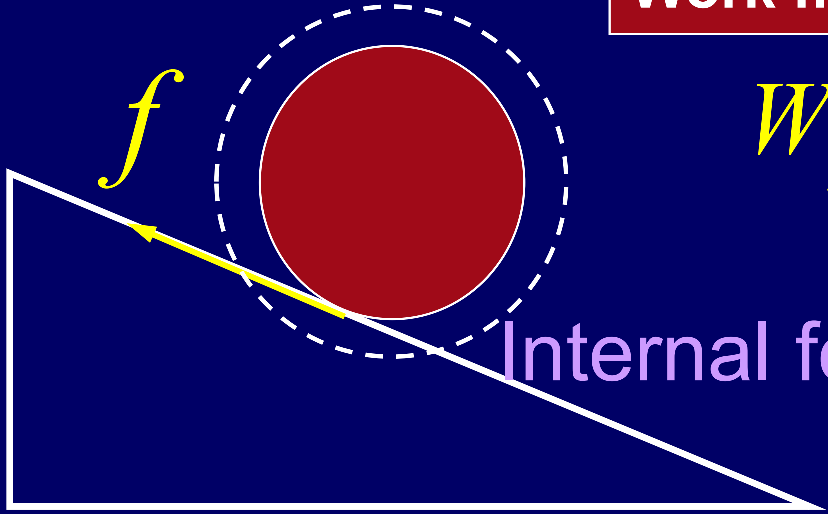
Center-of-mass energy

The center-of-mass energy equation (COM)

$$K = K_{cm} + K_R \qquad W_{ext} = \Delta K_{cm} + \Delta K_R$$

$$dW_{ext} = \Sigma F_{ext,i}dx_i \neq F_{ext}dx_{cm} \qquad dW_{ext,\mathrm{int}} = \Sigma F_{ext,\mathrm{int}}dx_i$$

Work from friction force



$$W_f = ? \quad 0 \neq \vec{f} \cdot d\vec{s} = -\Delta K_{cm}$$

$$\vec{\tau}_f \cdot d\vec{\theta} = fRd\theta = \Delta K_R$$

Rotation without slipping

$$ds = Rd\theta \qquad \Sigma W_f = 0$$

Rotation with slipping

$$ds \neq Rd\theta \qquad \Sigma W_f \neq 0$$

$$\Sigma W_f = ?$$

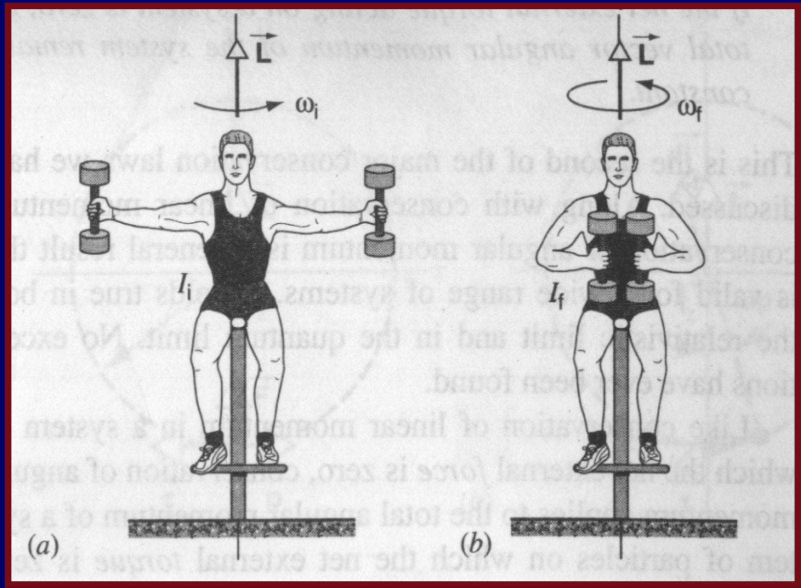
$$W_f = \Sigma \vec{f}_i \cdot d\vec{s}_i = \Delta E_{\text{int}}$$

$$\Delta E_{\text{int}} + \Delta K + \Delta U = 0$$

$$\Delta E_{\text{int}} = 0 \qquad \Delta K + \Delta U = 0$$

$$\Delta E_{\text{int}} > 0 \qquad \Delta K + \Delta U < 0$$

$$\Delta E_{\text{int}} < 0 \qquad \Delta K + \Delta U > 0$$



$$K = \frac{1}{2} I\omega^2 = 2270J$$

$$K' = \frac{1}{2} I'\omega'^2 = 11250J \approx 5K$$

$$W_{ext} = 0$$

$$I = 60 \times 0.15^2 + 2 \times 20 \times 0.5^2 = 11.35$$

$$\Delta K + \Delta E_{\text{int}} = 0$$

$$I' = (60+40) \times 0.15^2 = 2.25$$

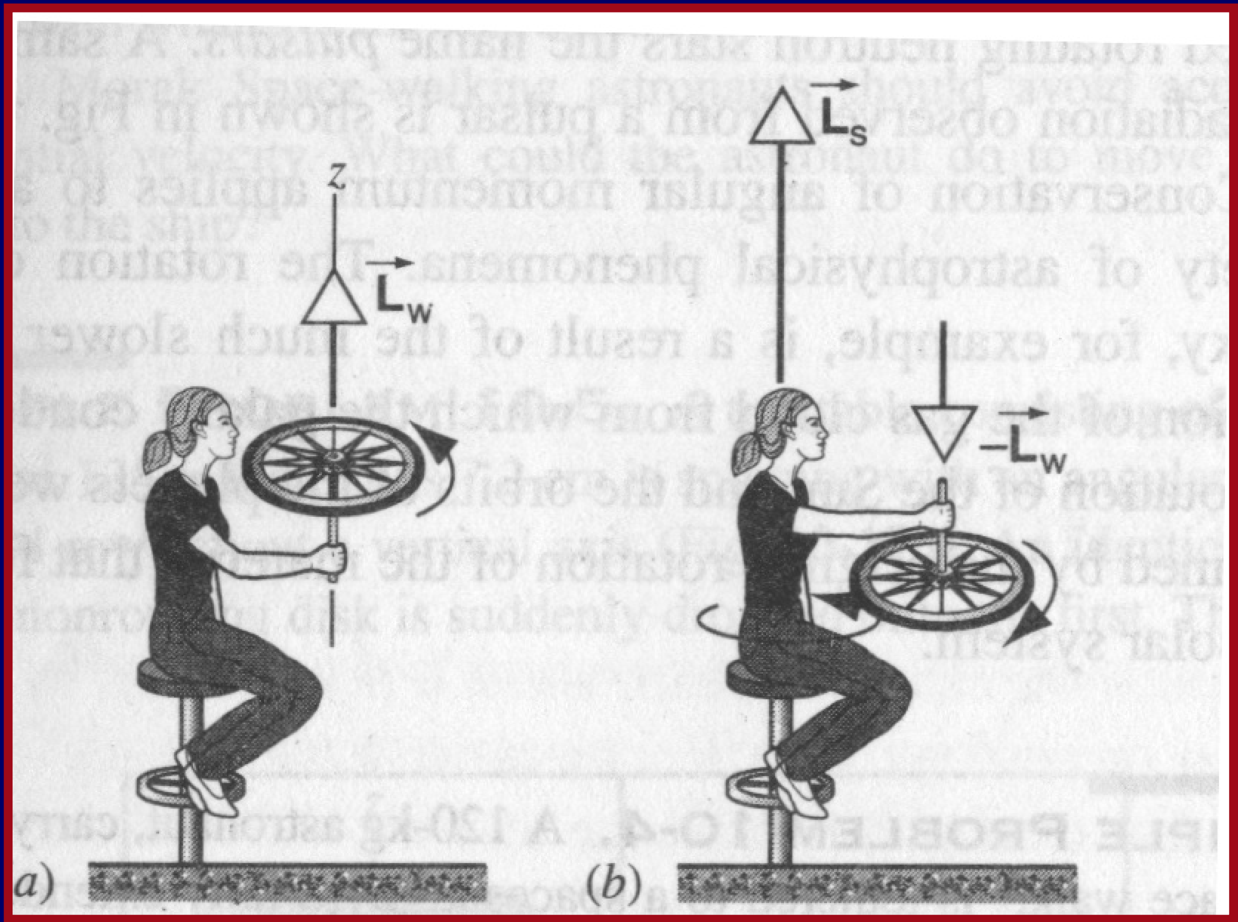
$$\Delta E_{\text{int}} < 0$$

$$I\omega = I'\omega'$$

$$\omega = 20/s \qquad \omega' = 100/s$$

$$\Delta K > 0$$

$$\Delta L = 0 \quad L_w = L_s - L_w \quad L_s = 2L_w \quad K_i = \frac{1}{2} I\omega^2$$



$$K_f = \frac{1}{2} I\omega_s^2 + \frac{1}{2} I\omega^2$$

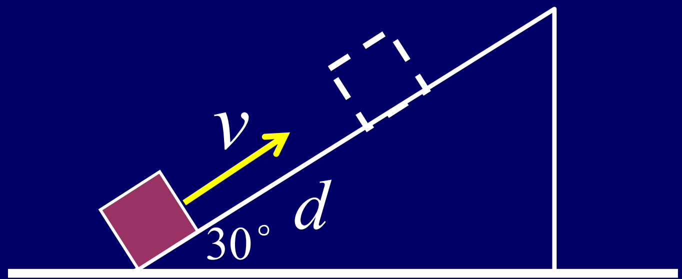
$$\Delta K > 0$$

$$W_{ext} = 0$$

$$\Delta K + \Delta E_{\text{int}} = 0$$

$$\Delta E_{\text{int}} < 0$$

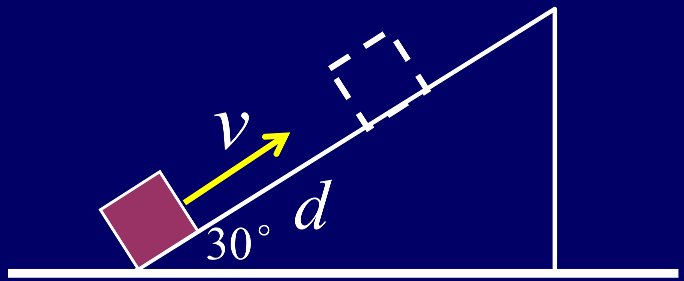
Example



A block with mass *m* and initial speed *v*, its speed gradually decreases to zero when it travel a distance *d*

Find (1) ΔE_{int} for the system of block + table+Earth due to friction. (2) If the block slide from the rest back down the table, find the speed when the block passes through the initial point.

Example



$$\Delta U = mgh = mgd\sin 30^\circ = \frac{mgd}{2}$$

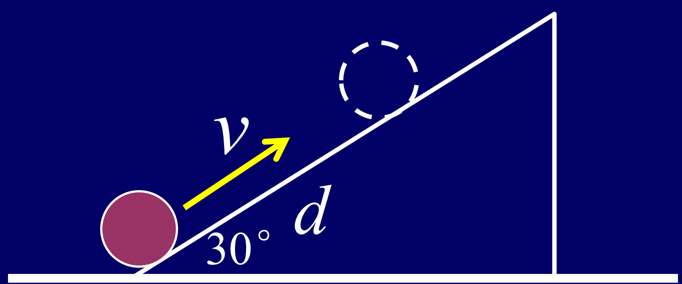
$$\Delta K = -\frac{1}{2}mv^2 \qquad \Delta U + \Delta K + \Delta E_{\text{int}} = 0$$

$$\Delta E_{\text{int}} = -(\Delta U + \Delta K) = \frac{1}{2}mv^2 - \frac{mgd}{2}$$

$$\Delta U = 0 \qquad \Delta K = \frac{1}{2}mv_t^2 - \frac{1}{2}mv^2 \qquad \Delta K + 2\Delta E_{\text{int}} = 0$$

$$\frac{1}{2}mv_t^2 - \frac{1}{2}mv^2 - mgd + mv^2 = 0 \qquad v_t = \sqrt{2gd - v^2}$$

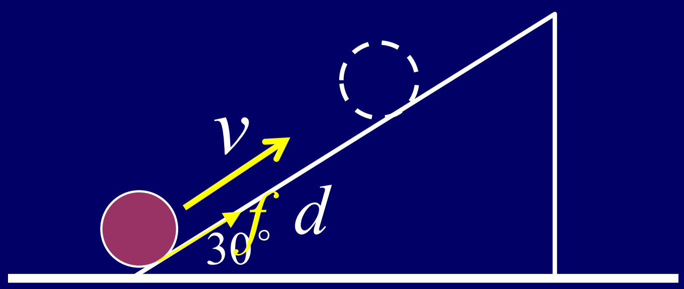
Example



A cylinder with mass m and initial speed v, it rolls without slipping, its speed gradually decreases to zero when it travel a distance d

Find (1) ΔE_{int} for the system of cylinder + table+Earth due to friction. (2) If cylinder rolls from the rest back down the table, find the speed when it passes through the initial point.

Example



$$f - mg\sin\theta = ma$$

$$-fR = \frac{1}{2}mR^2\alpha$$

$$-f = \frac{1}{2}mR\alpha = \frac{1}{2}ma$$

$$-mg\sin\theta = \frac{3}{2}ma$$

$$a = -\frac{2g\sin\theta}{3}$$

$$0 - v^2 = 2ad$$

$$v^2 = \frac{4gd\sin\theta}{3} = \frac{4gh}{3}$$

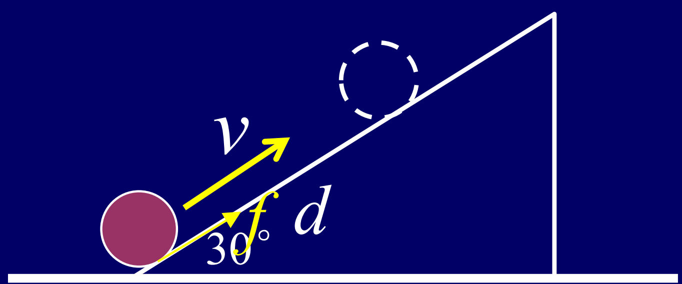
$$\Delta E_{\text{int}} = -(\Delta U + \Delta K)$$

$$\Delta E_{\text{int}} = mgh - \frac{1}{2}mv^2 - \frac{1}{2}I\omega^2$$

$$= mgh - \frac{1}{2}mv^2 - \frac{1}{2}\frac{1}{2}mR^2\omega^2$$

$$= mgh - \frac{3}{4}mv^2 = 0$$

Example



$$mg\sin\theta - f = ma$$

$$fR = \frac{1}{2}mR^2\alpha$$

$$f = \frac{1}{2}mR\alpha \quad = \frac{1}{2}ma$$

$$mg\sin\theta = \frac{3}{2}ma$$

$$a = \frac{2g\sin\theta}{3}$$

$$v_t^2 - 0 = 2ad$$

$$v_t^2 = \frac{4gd\sin\theta}{3} = \frac{4gh}{3}$$

$$r << R$$

$$v_{cm} = ?$$

$$F = ?$$

$$2mgR = \frac{7}{10} m v_{cm}^2$$

$$v_{cm} = r\omega \qquad v_{cm} = \sqrt{\frac{20gR}{7}}$$

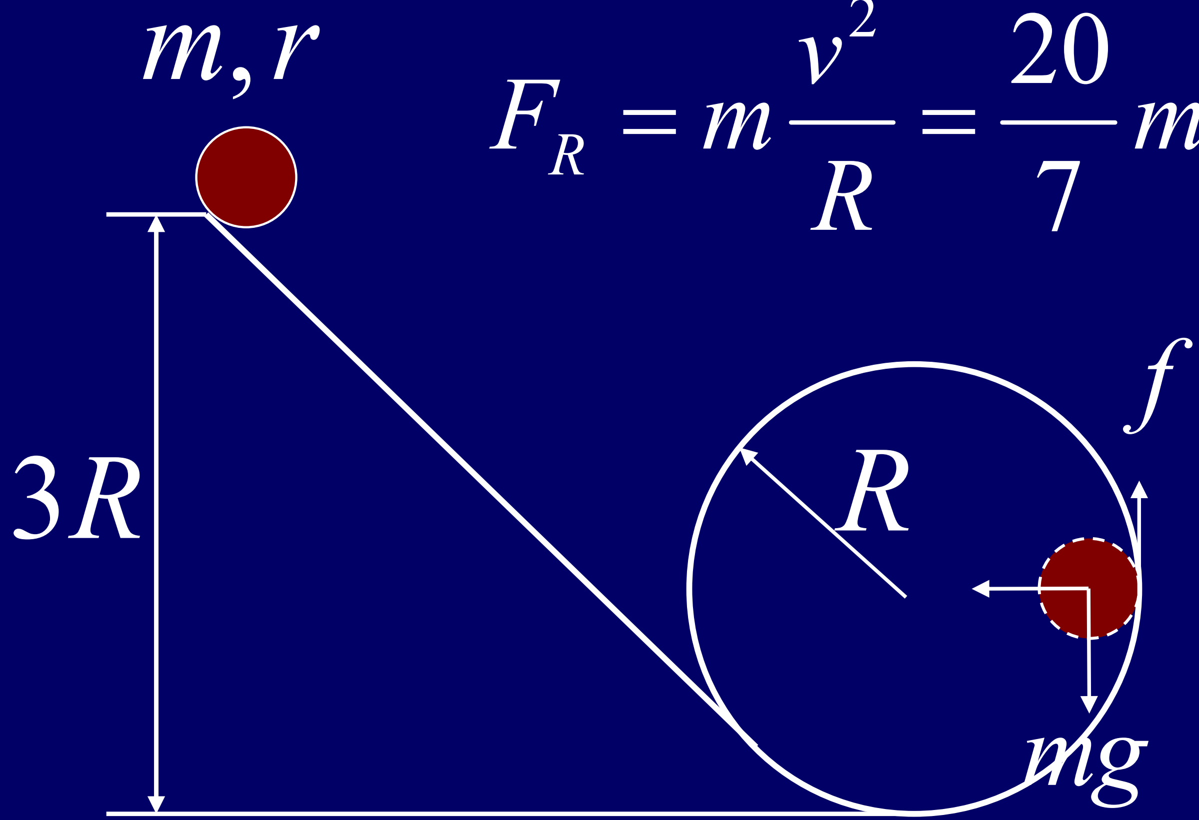
$$F_R = m\frac{v^2}{R} = \frac{20}{7}mg$$

$$mg - f = ma_{cm}$$

$$f = \frac{2}{5}ma_{cm}$$

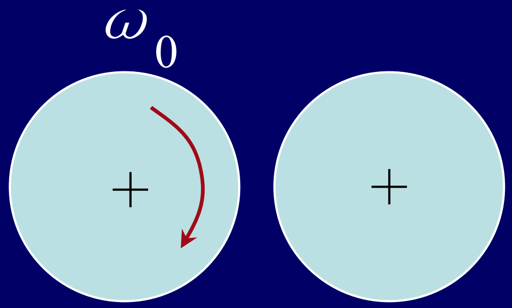
$$a_{cm} = r\alpha$$

$$f = \frac{2}{7}mg$$

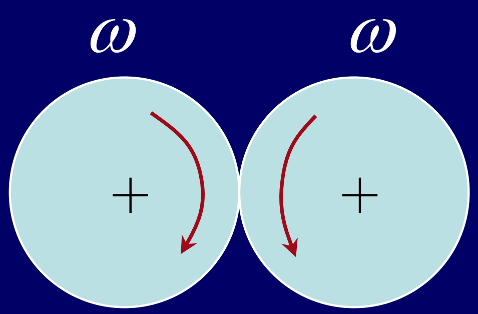


Rolling without slipping

Example



Example



for each wheel $\Delta E_{\text{int}} = ?$

$$K = \frac{1}{2} I\omega_0^2$$

$$\omega = \omega_0 - \alpha t$$

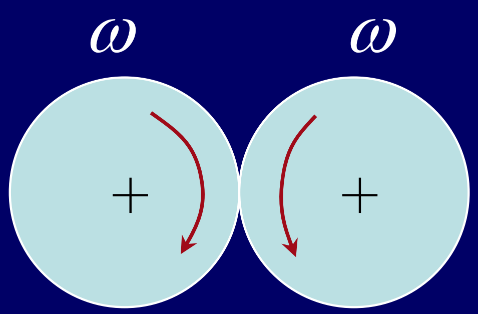
$$K' = 2 \times \frac{1}{2} I\omega^2 = I\left(\frac{\omega_0}{2}\right)^2 = \frac{1}{4} I\omega_0^2$$

$$\omega = \alpha t$$

$$K - K' = \frac{1}{4} I\omega_0^2 = \Delta E_{\text{int}}$$

$$\omega = \frac{\omega_0}{2}$$

$$\frac{1}{2}\Delta E_{\text{int}} = \frac{1}{8} I\omega_0^2$$



Example

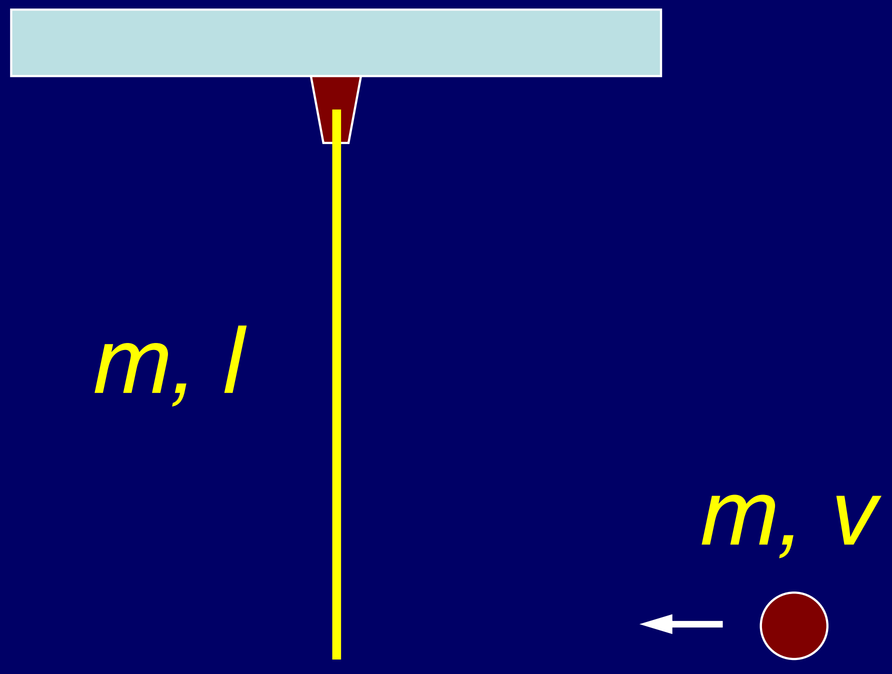
for each wheel $\Delta E_{\text{int}} = ?$

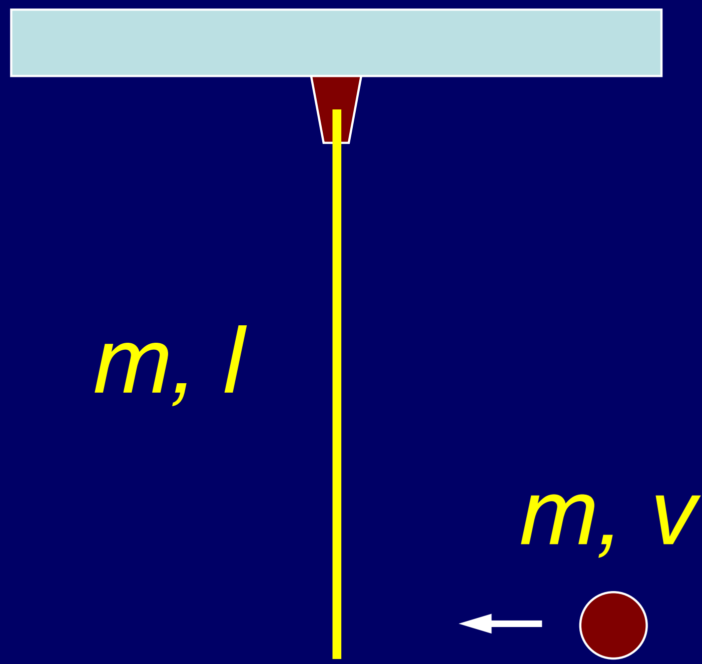
$$K = \frac{1}{2} I \omega_0^2$$

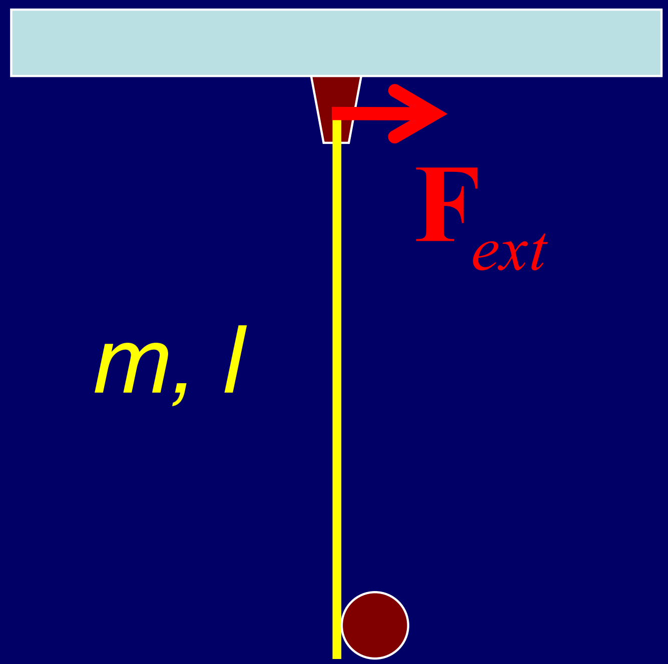
$$K' = 2 \times \frac{1}{2} I \omega^2 = I\left(\frac{\omega_0}{2}\right)^2 = \frac{1}{4} I \omega_0^2$$

$$K - K' = \frac{1}{4} I \omega_0^2 = \Delta E_{\text{int}}$$

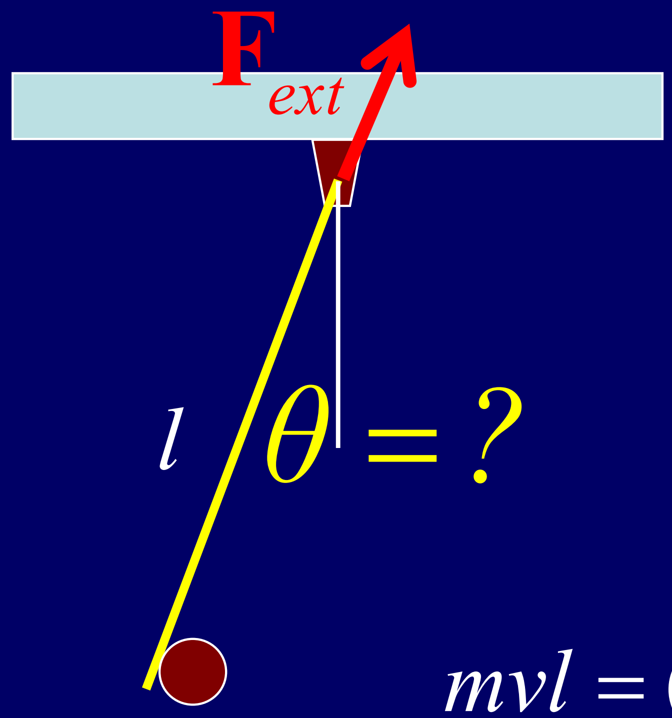
$$\frac{1}{2} \Delta E_{\text{int}} = \frac{1}{8} I \omega_0^2$$


m, l
m, v


m, l
m, v



$\mathbf{F}_{ext}$
m, l



During collision

$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0 \qquad \Delta K + \Delta U \neq 0$$

$$\Sigma \tau_{ext} = 0, \quad \Delta\vec{L} = 0$$

$$L_0 = mvl \qquad L' = (ml^2 + \frac{1}{3}ml^2)\omega$$

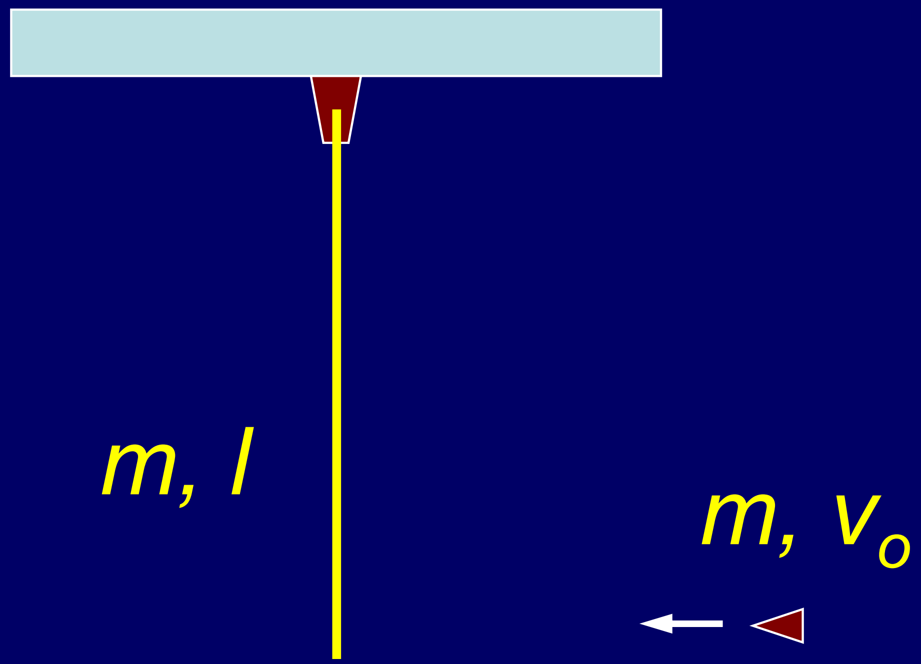
$$mvl = (ml^2 + \frac{1}{3}ml^2)\omega \qquad \omega = \frac{3v}{4l}$$

After collision

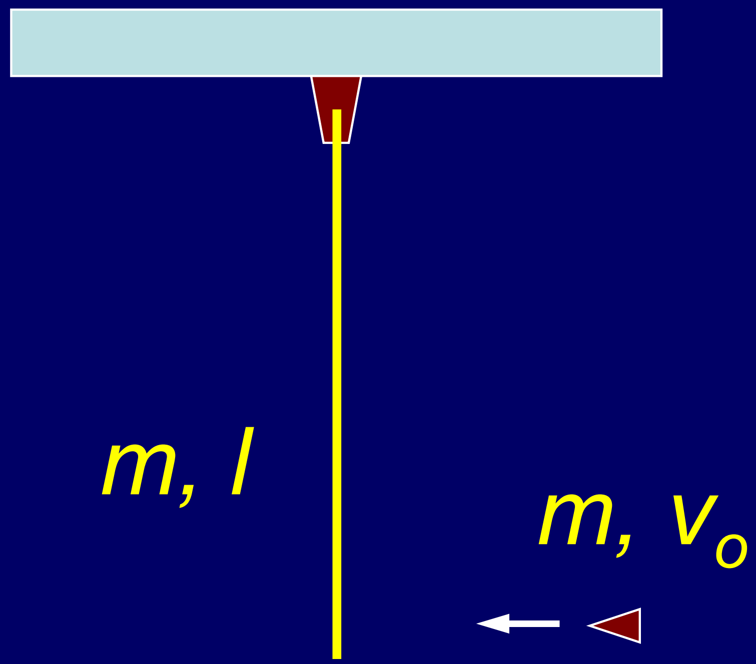
$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0 \qquad \Sigma \tau_{ext} \neq 0, \quad \Delta\vec{L} \neq 0$$

$$\Delta K + \Delta U = 0 \qquad \frac{1}{2}\frac{4}{3}ml^2\omega^2 = \frac{3}{2}mgl(1 - \cos\theta)$$

$$\cos\theta = 1 - \frac{v^2}{4gl}$$

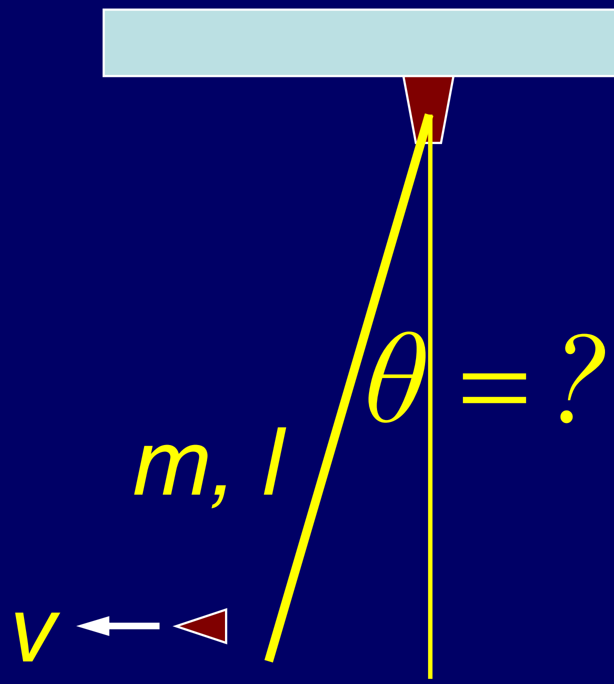


m, l
m, vo


m, l
m, v_o

During collision

$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0$$
$$\Delta K + \Delta U \neq 0$$
$$\Sigma \tau_{ext} = 0, \quad \Delta\vec{L} = 0$$

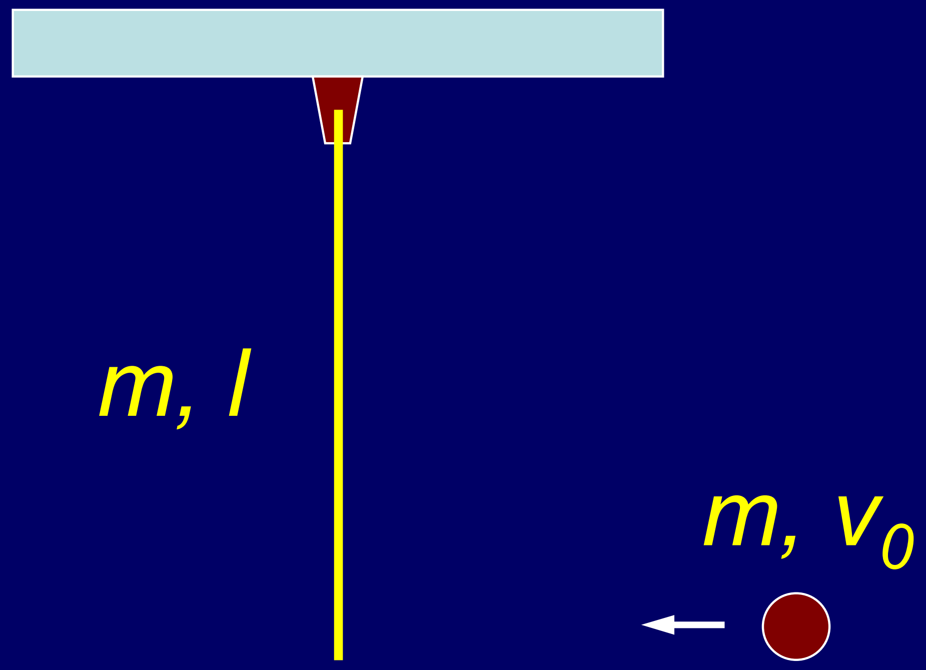


$$mv_0 l = I\omega + mvl$$
$$mv_0 l = \frac{1}{3}ml^2\omega + mvl$$
$$\omega = \frac{3(v_0 - v)}{l}$$

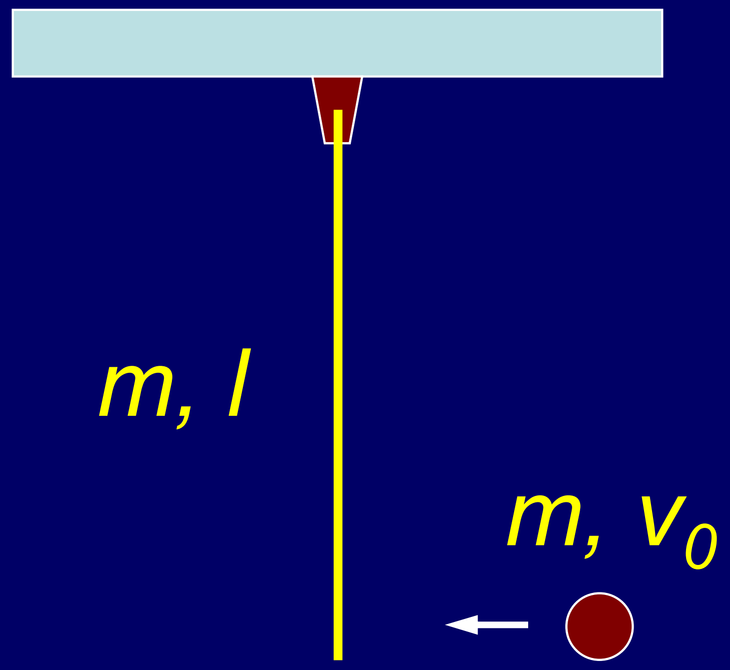
After collision

$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0$$
$$\Sigma \tau_{ext} \neq 0, \quad \Delta\vec{L} \neq 0$$
$$\Delta K + \Delta U = 0$$

$$\frac{1}{2}I\omega^2 = \frac{1}{2}mgl(1-\cos\theta)$$
$$\cos\theta = 1 - \frac{3(v_0 - v)^2}{gl}$$



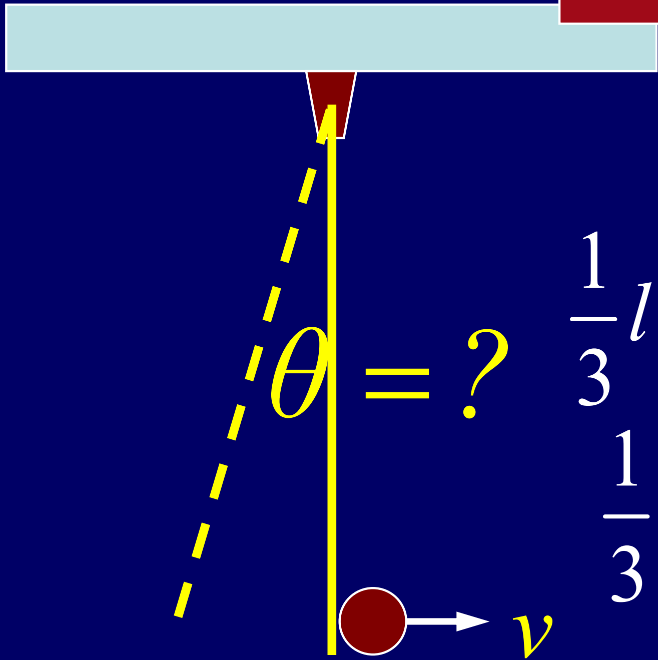
m, l
m, v0



m, l
m, v0

Elastic collision

$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0$$
$$\Delta K + \Delta U = 0$$
$$\Sigma \tau_{ext} = 0, \quad \Delta\vec{L} = 0$$



$$\frac{1}{3} l^2\omega^2 = v_0^2 - v^2$$

$$\frac{1}{3} l\omega = v_0 + v$$

$$l\omega = v_0 - v$$

$$\omega = \frac{3v_0}{2l}$$

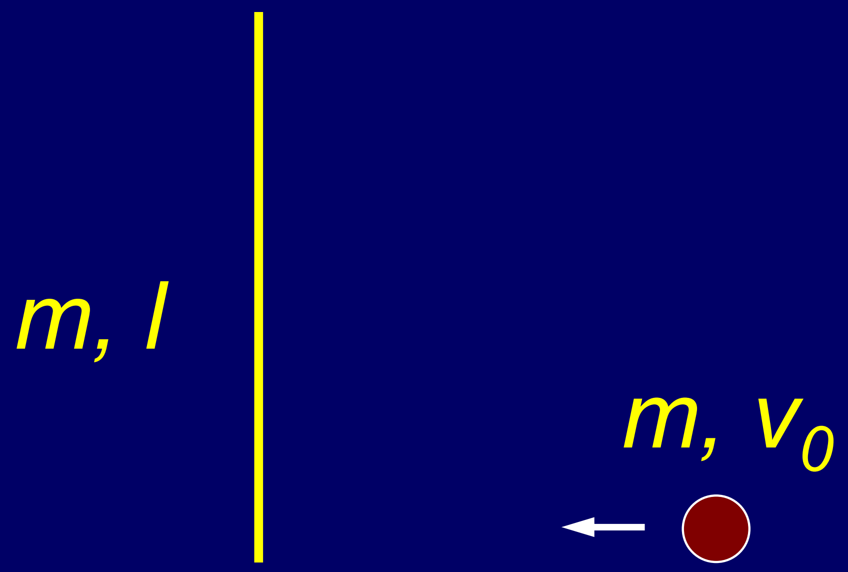
$$v = -\frac{1}{2} v_0$$

After collision

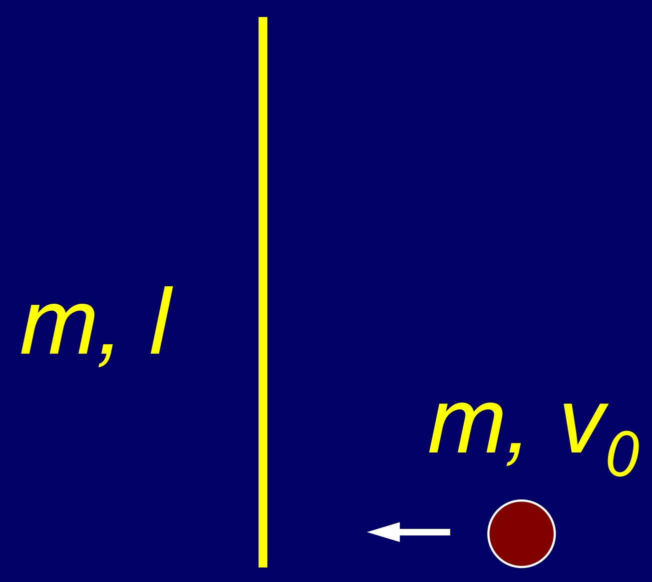
$$\Sigma F_{ext} \neq 0, \quad \Delta\vec{p} \neq 0$$
$$\Sigma \tau_{ext} \neq 0, \quad \Delta\vec{L} \neq 0$$
$$\Delta K + \Delta U = 0$$

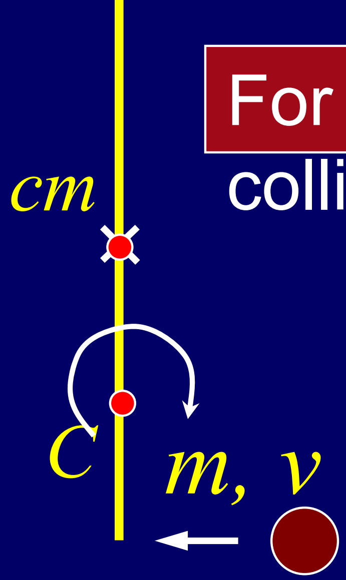
$$\frac{1}{3} l\omega^2 = g(1 - \cos\theta)$$

$$\cos\theta = 1 - \frac{3v_0^2}{4gl}$$



m, l
m, v_0


m, l
m, v0

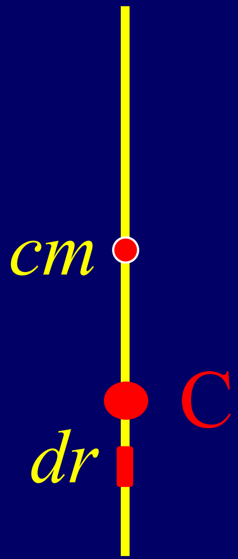


For elastic collision:

$$\frac{1}{2}mv^2 = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}mv_{cm}{}^2 + \frac{1}{2}mv_f{}^2$$

$$\Sigma F_{ext} = 0, \quad \Delta\vec{p} = 0 \qquad mv = mv_{cm} + mv_f$$

$$\Sigma\tau_{ext} = 0, \quad \Delta\vec{L} = 0$$



about cm $$d\vec{p} = \lambda dr(\vec{v}_{cm} + \vec{\omega}\times\vec{r})$$

about C $$d\vec{p} = \lambda dr(\vec{v}_{cm} + \vec{\omega}\times\vec{r}_{cm,C} + \vec{\omega}\times\vec{r})$$

about cm $$\frac{mvl}{2} = \frac{mv_f l}{2} + \left|\int_{-l/2}^{l/2}\vec{r}\times d\vec{p}\right| = \frac{mv_f l}{2} + I_{cm}\omega$$

about C $$\frac{mvl}{4} = \frac{mv_f l}{4} + \left|\int_{-l/4}^{3l/4}\vec{r}\times d\vec{p}\right| = \frac{mv_f l}{4} - \frac{mv_{cm} l}{4} - \frac{1}{16}ml^2\omega + I_C\omega$$

about cm $\left|\int_{-l/2}^{l/2} \vec{r}\times(\omega\times\vec{r})\lambda dr\right| = I_{cm}\omega$ $I_{cm} = ml^2/12$

about C $\left|\int_{-l/4}^{3l/4} \vec{r}\times(\omega\times\vec{r})\lambda dr\right| = I_C\omega$ $I_C = 7ml^2/48$

about cm $$\frac{mvl}{2} = \frac{mv_f l}{2} + \left|\int_{-l/2}^{l/2} \vec{r}\times d\vec{p}\right| = \frac{mv_f l}{2} + \frac{1}{12}ml^2\omega$$

about C $$\frac{mvl}{4} = \frac{mv_f l}{4} + \left|\int_{-l/4}^{3l/4} \vec{r}\times d\vec{p}\right| = \frac{mv_f l}{4} - \frac{mv_{cm} l}{4} + \frac{1}{12}ml^2\omega$$

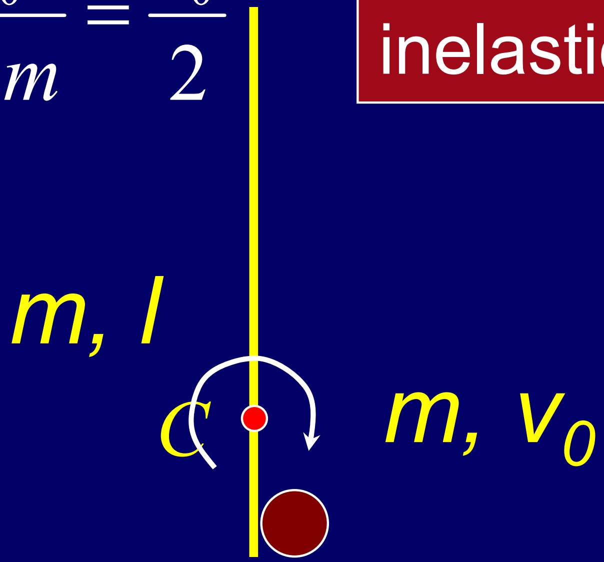
$$v = v_f + v_{cm}$$

Final solutions:

$v_f = 3v/5$ $v_{cm} = 2v/5$ $\omega = 12v/5l$

During collision

$$v_{cm} = \frac{mv_0}{m+m} = \frac{v_0}{2}$$



inelastic collision

$$\Sigma \tau_{ext} = 0, \quad \Delta \vec{L} = 0$$

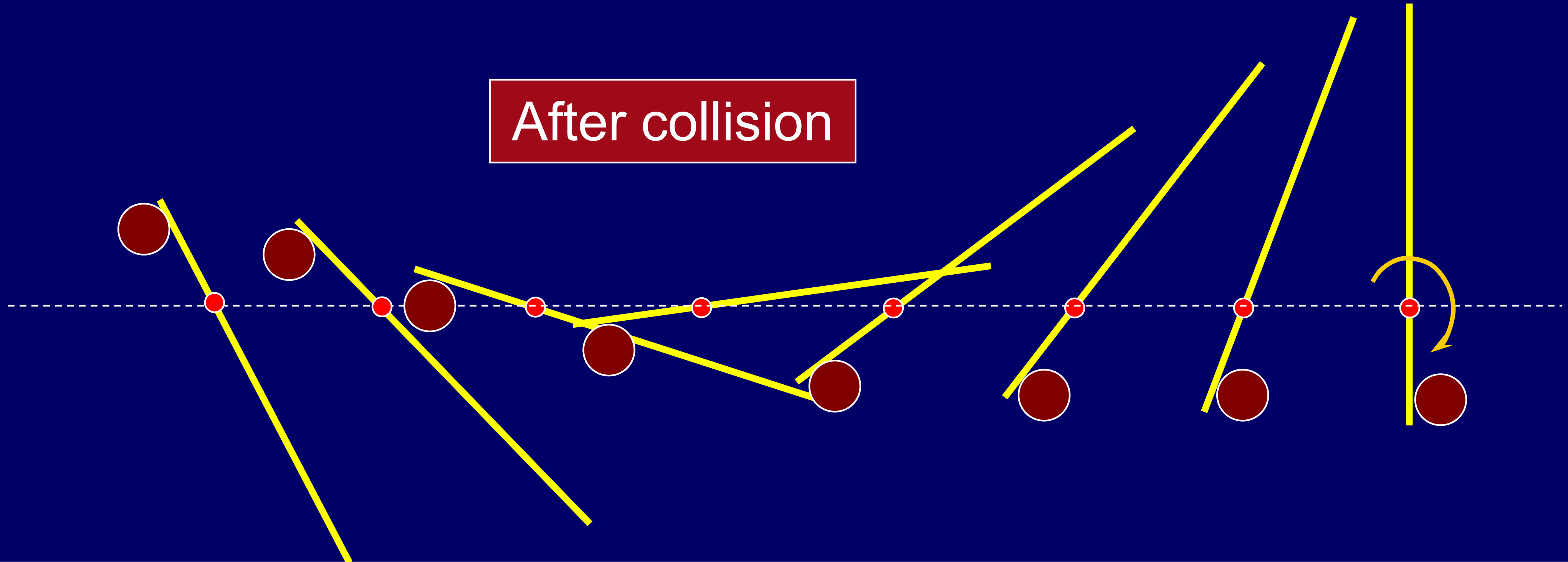
$$\Sigma F_{ext} = 0, \quad \Delta \vec{p} = 0$$

$$\Delta K + \Delta U \neq 0$$

$$\frac{mv_0 l}{4} = I\omega = \frac{5ml^2}{24}\omega \qquad \omega = \frac{6v_0}{5l}$$

$$I = \frac{ml^2}{16} + \left(\frac{ml^2}{12} + \frac{ml^2}{16}\right) = \frac{5ml^2}{24}$$

After collision



Review

$$if\ \Sigma F_{ext} = 0, \quad \Delta\vec{p} = 0$$

$$if\ \Sigma \tau_{ext} = 0, \quad \Delta\vec{L} = 0$$

$$if\ W_{ext} = 0, \quad \Delta E = 0$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$if\ \Delta E_{\text{int}} = 0\,, \ \Delta K + \Delta U = 0$$

CHAP.13

Exercises

P296 : 14, 15, 19

Problems

P297: 1, 2, 5